

Azal Ahmad Khan

✉ khan1069@umn.edu  [azalahmadkhan.github.io](https://github.com/azalahmadkhan)  [Google Scholar](https://scholar.google.com/citations?user=...)  +1(763)485-1561

Education

University of Minnesota, Twin-Cities Fall 2024 – Present
Ph.D., Computer Science and Engineering (GPA: 3.95/4.0)

- › Advisor: [Ali Anwar](#)
- › Amazon ML Systems Fellow (2025) & GAGE Fellow (2024)

Indian Institute of Technology (IIT), Guwahati Fall 2020 – Spring 2024
B.Tech., Chemical Science and Technology

Research Experience

NEC Laboratories America, San Jose, CA Summer 2025
Research Intern with [Vijay Kumar](#) & [Manmohan Chandrakar](#)

University of Minnesota, Twin-Cities, MN Fall 2024 - Present
Graduate Assistant with [Ali Anwar](#)

University of Minnesota Spring 2022 – Spring 2024
Research Intern with [Ali Anwar](#)

University of New South Wales Spring 2022 – Fall 2023
Research Intern with [Rohitash Chandra](#)

Awards & Grants

2025 [Amazon Machine Learning System Fellowship.](#)
2024 [University of Minnesota GAGE Fellowship.](#)
2024 [Global Undergraduate Research Award](#), Highly Commended.
2024 Travel Grant, [Google Research Week](#).

Publications & Pre-prints

* Equal Contribution

Under review [Optimizing Edge-Case Discovery in Autonomous Vehicle Testing via LLMs](#)
Azal Ahmad Khan, Vijay Kumar BG, Sparsh Garg, Ali Anwar, Manmohan Chandrakar.
Under review .

arXiv [Retrieval-of-Thought: Efficient Reasoning via Reusing Thoughts](#)
Azal Ahmad Khan*, Ammar Ahmed*, Ayaan Ahmad, Sheng Di, Zirui Liu, Ali Anwar.
Pre-print.

NeurIPS '25 [Beyond Expectations: Quantile-Guided Alignment for Risk-Calibrated Language Models](#)
Xinran Wang, Jin Du, **Azal Ahmad Khan**, Qi Le, Enmao Diao, Jiawei Zhou, Jie Ding, Ali Anwar.

EMNLP '25 [Accelerating LLM Reasoning via Early Rejection with Partial Reward Modeling](#)
Azal Ahmad Khan*, Seyyed Saeid Cheshmi*, Xinran Wang, Zirui Liu, Ali Anwar.
In Proceedings of Findings of EMNLP 2025.

arXiv [Sem-DPO: Mitigating Semantic Inconsistency in Preference Optimization for Prompt Engineering](#)

Azal Ahmad Khan*, Anas Mohamed*, Xinran Wang, Ahmad Faraz Khan, Shuwen Ge, Saman Bahzad Khan, Ayaan Ahmad, Ali Anwar.

Pre-print.

IROS '25 **Safety Aware Task Planning via Large Language Models in Robotics**

Azal Ahmad Khan*, Michael Andrew*, Muhammad Ali Murtaza, Sergio Aguilera, Rui Zhang, Jie Ding, Seth Hutchinson, Ali Anwar.

In the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025.

arXiv **LADs: Leveraging LLMs for AI-Driven DevOps**

Azal Ahmad Khan*, Ahmad Faraz Khan*, Anas Mohamed, Haider Ali, Suchithra Moolinti, Sabaat Haroon, Usman Tahir, Mattia Fazzini, Ali R. Butt, Ali Anwar.

Pre-print.

IPDPS '25 **IP-FL: Incentivized and Personalized Federated Learning**

Ahmad Faraz Khan, Xinran Wang, Qi Le, Zain ul Abdeen, **Azal Ahmad Khan**, Haider Ali, Ming Jin, Jie Ding, Ali Butt, Ali Anwar.

In Proceedings of the 39th IEEE International Parallel & Distributed Processing Symposium.

EuroSys '24 **FLOAT: Federated Learning Optimizations with Automated Tuning**

Ahmad Faraz Khan, **Azal Ahmad Khan**, Ahmed M Abdelmoniem, Samuel Fountain, Ali Butt, Ali Anwar.

In Proceedings of the Nineteenth European Conference on Computer Systems.

BigData '24 **Personalized Federated Learning Techniques: Empirical Analysis**

Azal Ahmad Khan, Ahmad Faraz Khan, Haider Ali, Ali Anwar.

In IEEE International Conference on Big Data 2024 (Short Paper).

BigData '24 **Mitigating Sycophancy in Large Language Models via Direct Preference Optimization**

Azal Ahmad Khan, Sayan Alam, Xinran Wang, Ahmad Faraz Khan, Debanga Raj Neog, and Ali Anwar.

In IEEE International Conference on Big Data 2024 (Short Paper).

Elsevier **A review of ensemble learning and data augmentation models for class imbalanced problems: combination, implementation and evaluation**

Azal Ahmad Khan, Omkar Chaudhari, Rohitash Chandra.

Expert Systems with Applications, Volume 244, 122778.

MDPI **A Quantum-Inspired Predator–Prey Algorithm for Real-Parameter Optimization**

Azal Ahmad Khan, Salman Hussain, Rohitash Chandra.

Algorithms 2024, 17(1), 33.

Technical Skills

Languages Python, C++ **Frameworks** PyTorch, TensorFlow, vLLM, Hugging Face
LLM Systems & Agents LangChain, AutoGen, Ollama, FastAPI
Tools Git, Docker, SQL, L^AT_EX